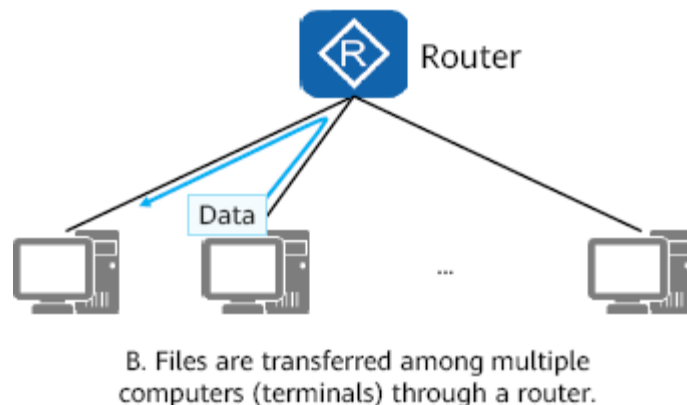
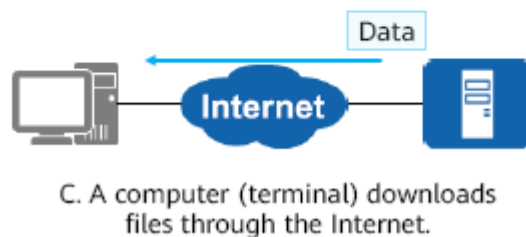
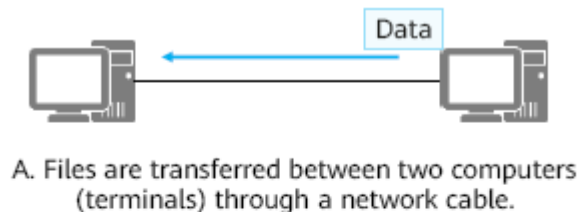




Communication and Networks

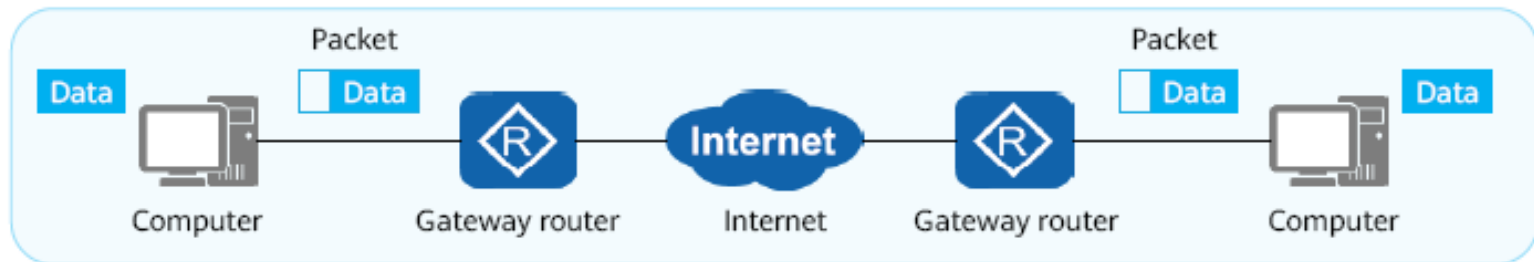
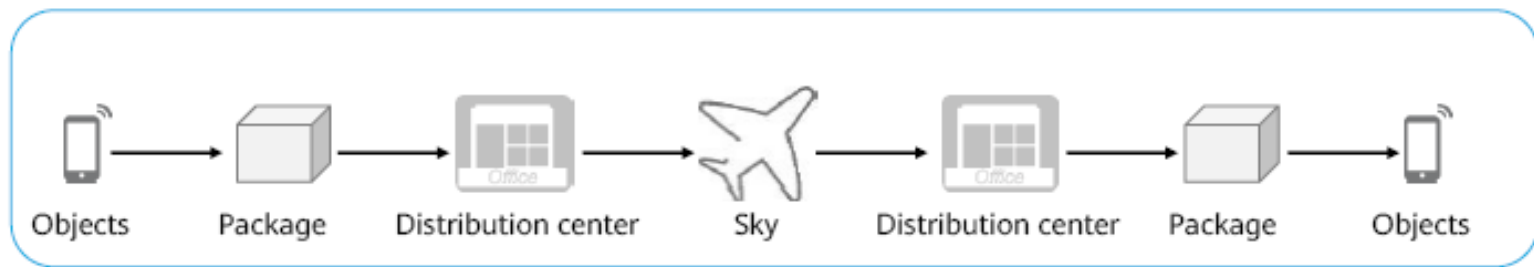
Concept of Network Communication

- Communication refers to the information transfer and exchange between people, between people and things, and between things through a certain medium and behavior.
- Network communication refers to communication between terminal devices through a computer network.
- Examples of network communication



Information Transfer Process

- Virtual information transfer is similar to real object transfer.



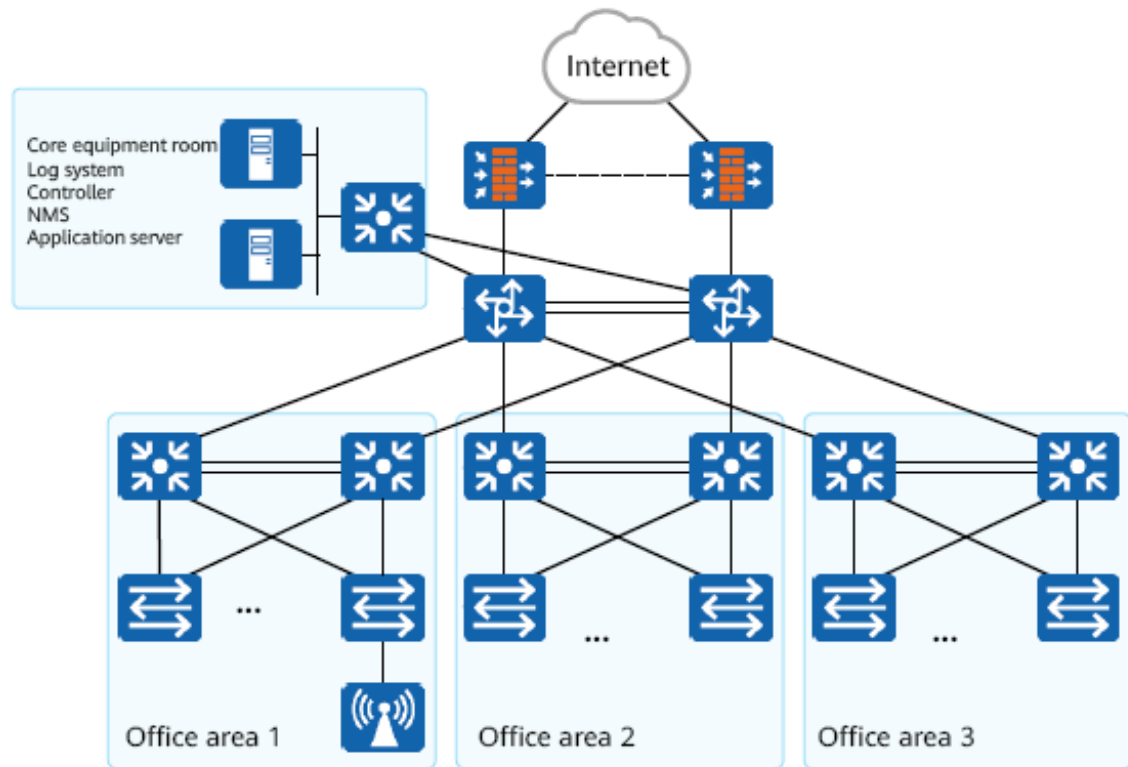
Common Terms

Term	Description
Data payload	Information conveyed
Packet	Data unit switched and transmitted on the network
Header	Information segment added before the data payload
Tail	Information segment added after the data payload
Encapsulation	Process of adding a header and a tail to a data payload to form a new packet
Decapsulation	Process of removing the header and tail from a packet to obtain the data payload
Gateway	Network device that provides functions such as protocol conversion, route selection, and data exchange
Router	Network device that selects a forwarding path for packets
Terminal device	End device of a data communication system, used as a sender or receiver of data

Concept of the Data Communication Network

Data Communication network:
Communication network that consists of routers, switches, firewalls, access controllers (ACs), access points (APs), PCs, network, printers, and servers.

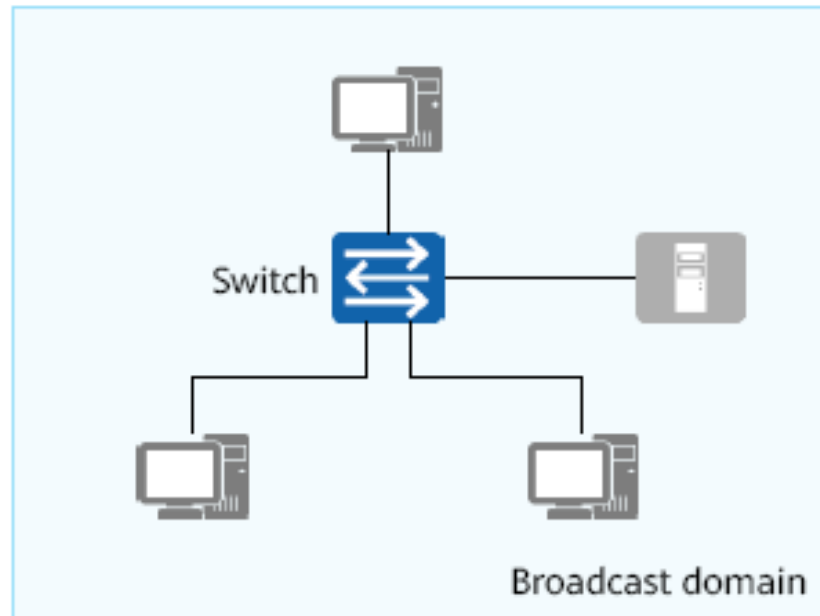
Function: To implement data communication



Switches

Switch: a device closest to end users, used to access the network and switch data frame.

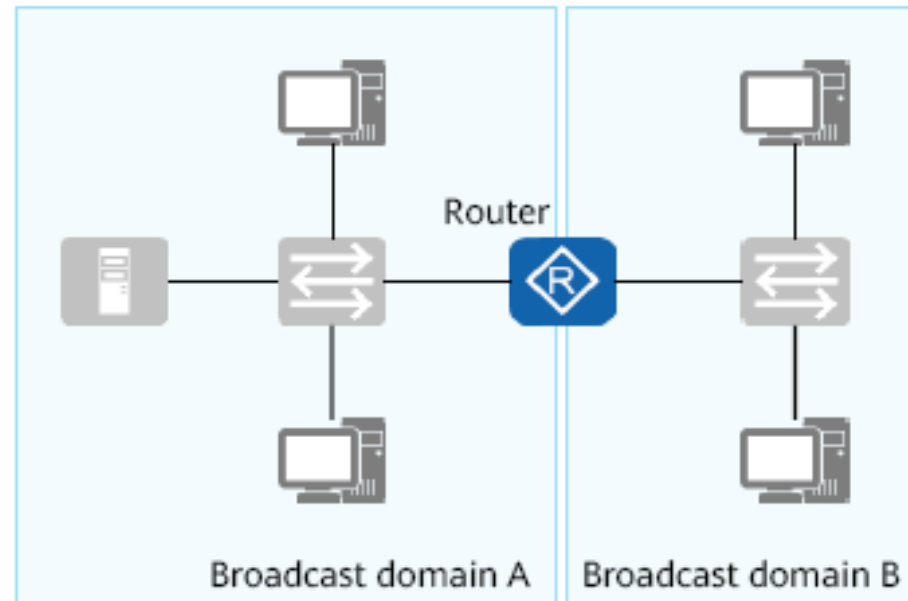
- Network access of terminals (such as PCs and servers).
- Layer 2 switching.



Router

Router: a network-layer device that forwards data packets on the Internet. Based on the destination address in a received packet, a router selects a path to send the packet to the next router or destination. The last router on the path is responsible for sending the packet to the destination host.

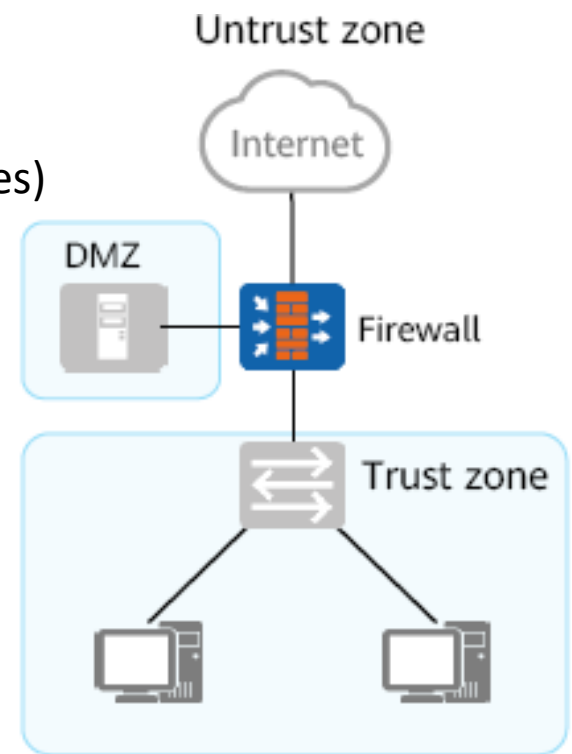
- Implementing communication between networks of the same type or different types
- Isolating broadcast domains
- Maintaining the routing table and running routing protocols
- Selecting routes and forwarding IP packets
- Implementing WAN access and network address translation
- Connecting Layer 2 networks established through switches



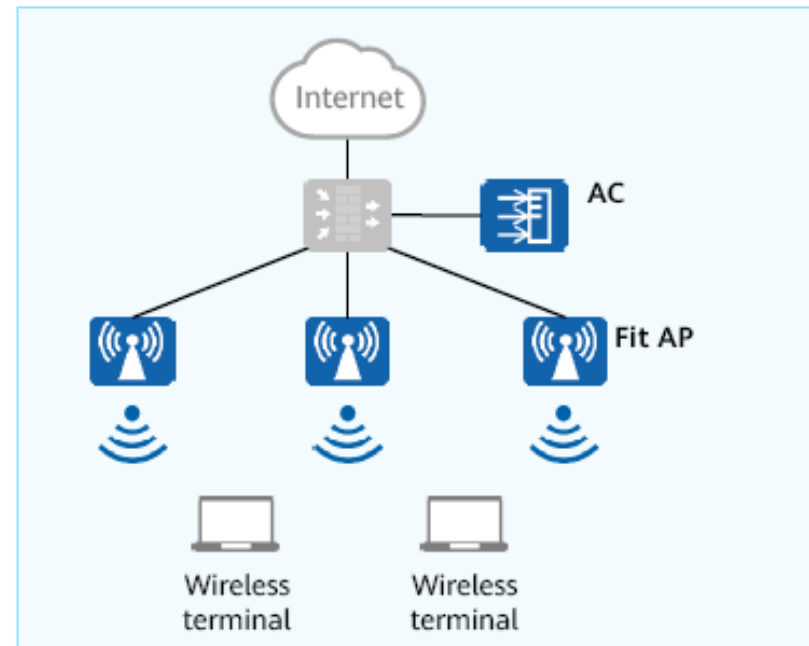
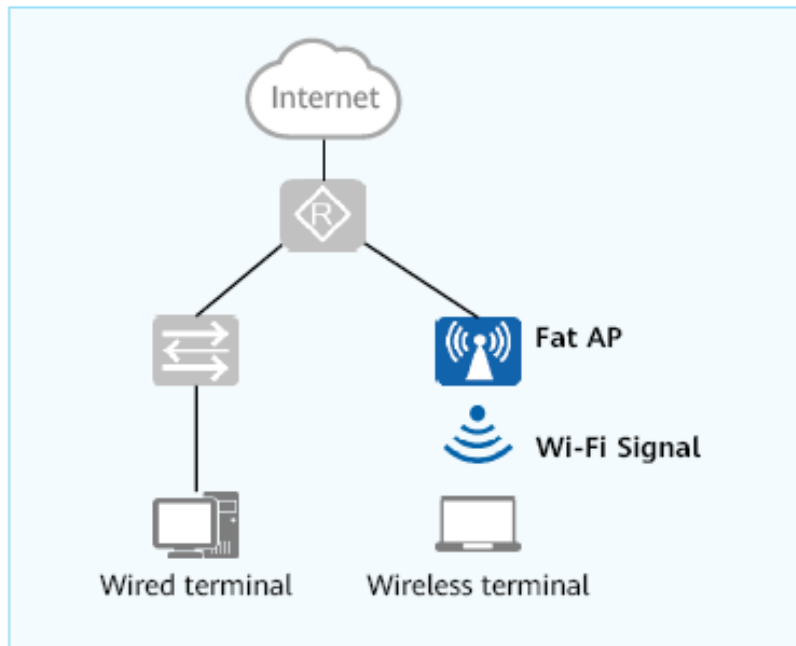
Firewall

Firewall: a network security device used to ensure secure communication between two networks. It monitors, restricts, and modifies data flows passing through it to shield information, structure, and running status of internal networks from the public network.

- Isolating networks of different security levels
- Implementing access control (using security policies) between networks of different security levels
- Implementing user identity authentication
- Implementing remote access
- Supporting data encryption and VPN services
- Implementing network address translation
- Implementing other security functions



Wireless devices



LAN, MAN, and WAN

Based on the geographical coverage, networks can be classified into local area networks (LANs), an area networks (MANs), and wide area networks (WANs).

- LAN
 - A LAN is a network that consists of computers, servers, and network devices in a geographic area. The coverage of generally within several thousand square meters.
 - Typical LANs include a company's office network, a cyber bar network, a home network.

LAN, MAN, and WAN

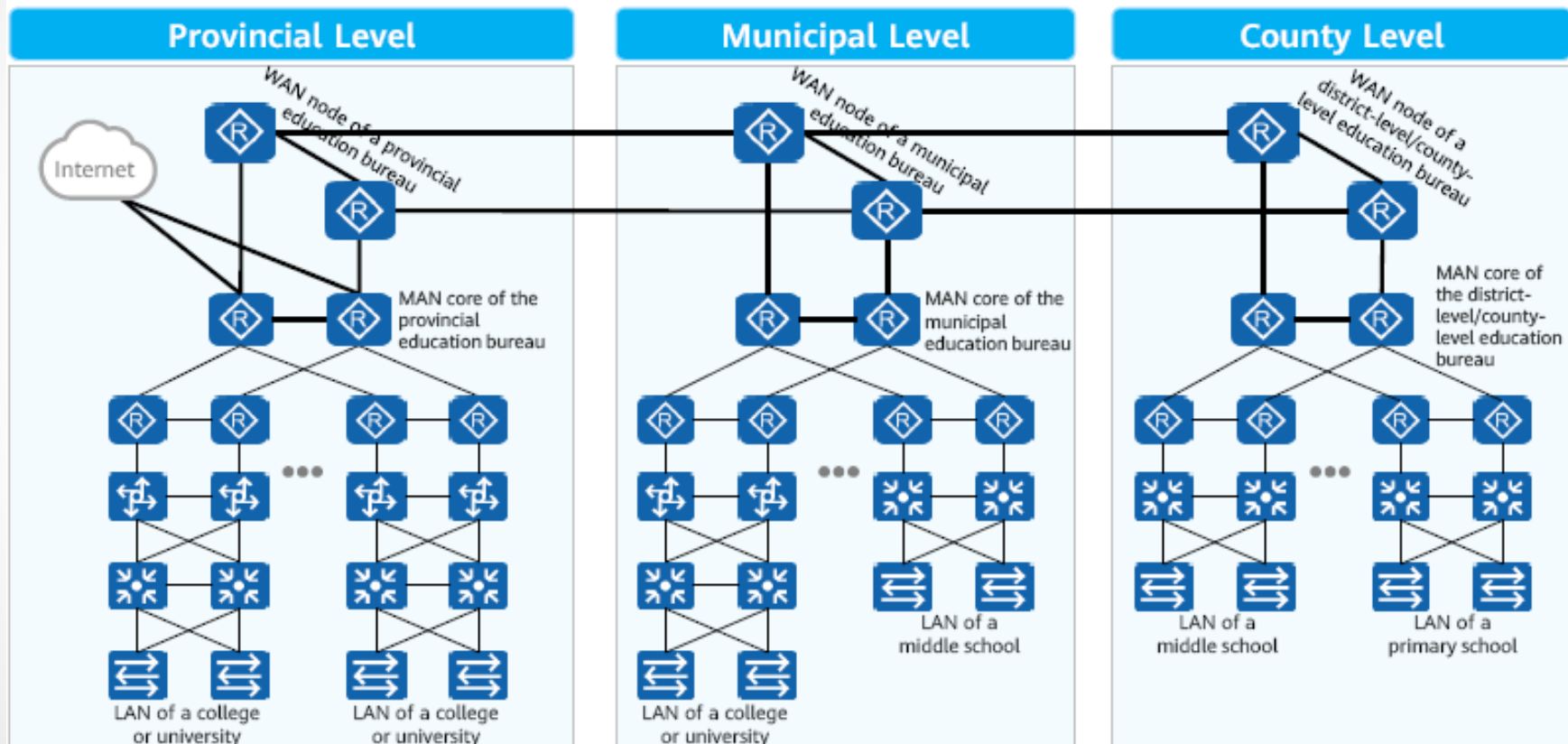
- MAN

- A MAN is a computer communication network established within a city.
- Typical MANs include broadband MANs, education MANs, and municipal or provincial -government private lines.

- WAN

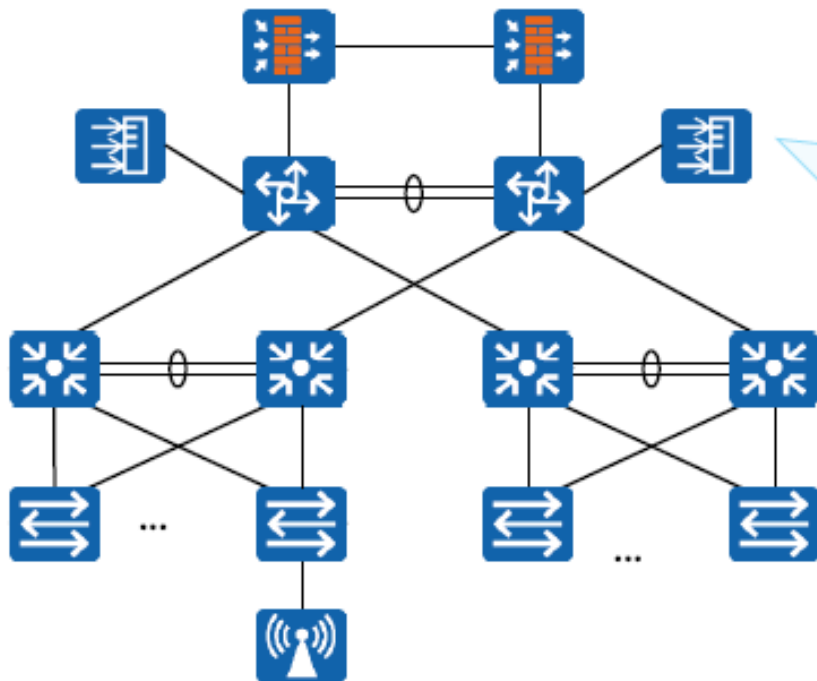
- A WAN generally covers a large geographical area ranging from tens of square kilometers to thousands of square kilometers. It can connect networks of multiple cities or even networks of countries (as an international large-scale network) and provide long-distance communication.
- The Internet is a typical WAN.

LAN, MAN, and WAN in the Education Industry



Network Topology

- A network topology is a structured layout presented using transmission media (such as twisted pairs and optical fibers) to interconnect various devices (such as computer terminals, routers, and switches).



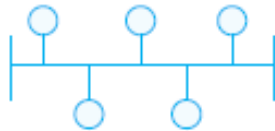
The network topology is used to describe the physical or logical structure of a network in the network engineering field, and is a very important network concept.

Network Topology Types

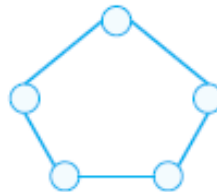
- Network topologies are classified into star, bus, ring, tree, , full-mesh, and partial-mesh network topologies.



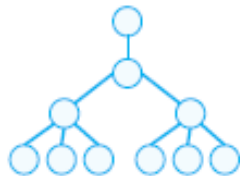
Star network topology



Bus network topology



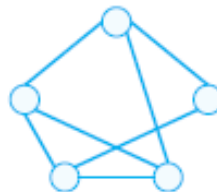
Ring network topology



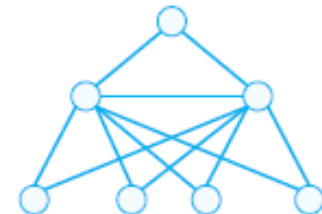
Tree network topology



Full-mesh network topology



Partial-mesh network topology



Combined network topology

Origin of the Story - Applications

- Applications are used to meet various requirements of people, such as web page access, online gaming, and online video playback.
- Information is generated along with applications. Texts, pictures, and videos are all information presentation modes.



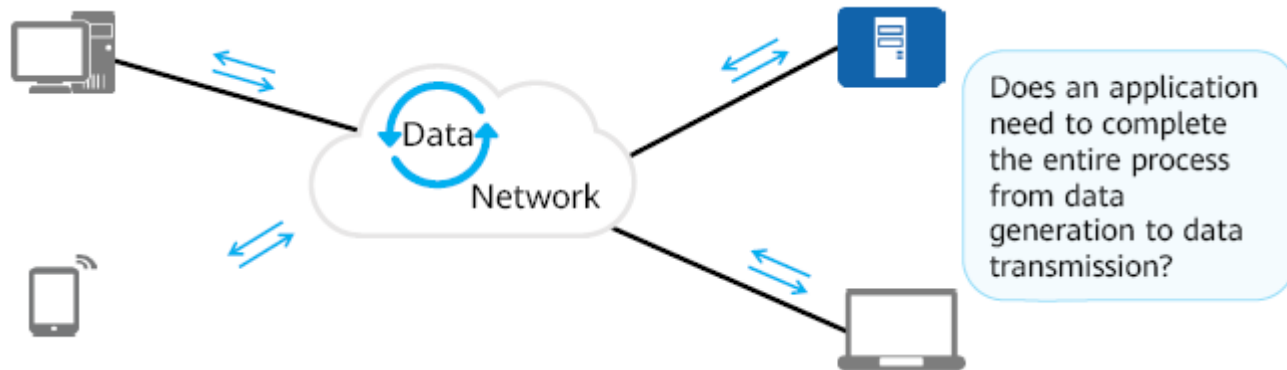
Application Implementation - Data

Data generation

- In the computer field, data is the carrier of all kinds of information.

Data transmission

- Data generated by most application needs to be transmitted between devices.

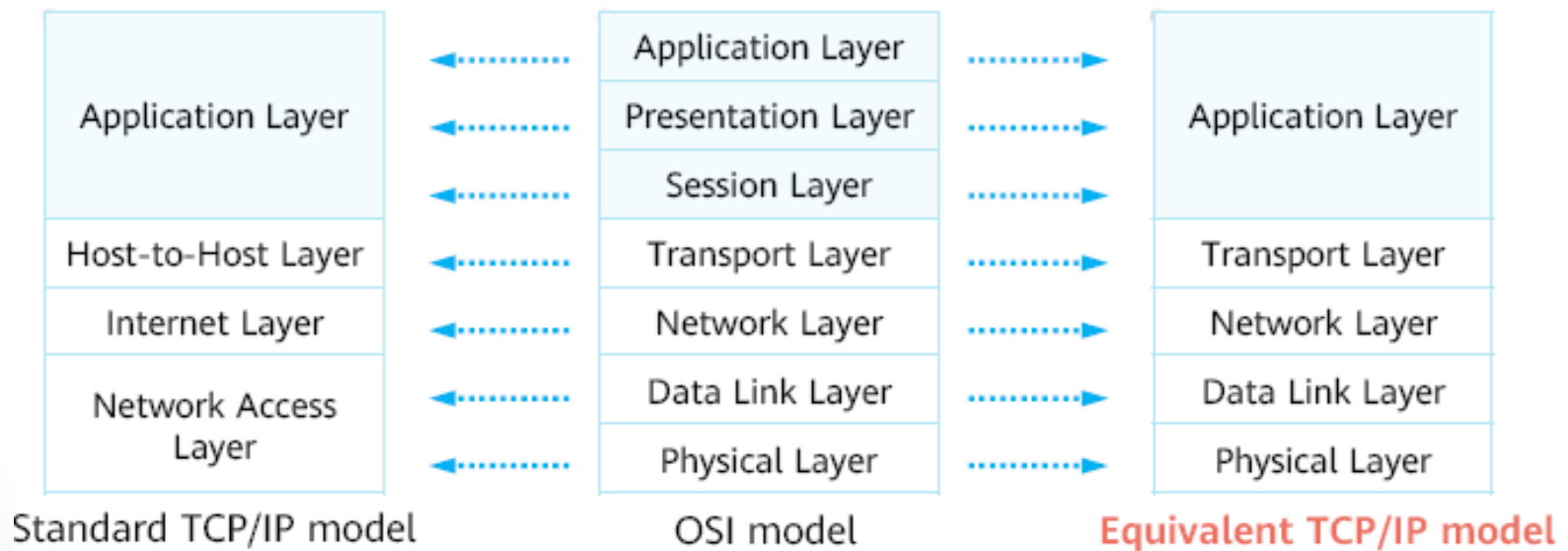


OSI Reference Model

7 Application Layer	Provides interfaces for applications
6 Presentation Layer	Translates data formats to ensure that the application-layer data of one system can be identified by the application layer of another system.
5 Session Layer	Establishes, manages, and terminates sessions between communicating parties
4 Transportation Layer	Establishes, maintains, and cancels an end-to-end data transmission process; controls transmission speeds and adjusts data sequences.
3 Network Layer	Defines logical addresses and transfers data from sources to destinations.
2 Data Link Layer	Encapsulates packets into frames, transmits frames in P2P or P2MP mode, and implements error checking.
1 Physical Layer	Transmits bitstreams over transmission media and defines electrical and physical specifications.

TCP/IP Reference Model

- The OSI protocol stack is complex, and the TCP and IP protocols are widely used in the industry. Therefore, the TCP/IP reference model becomes the mainstream reference model of the Internet.



Common TCP/IP protocol

- The TCP/IP protocol stack defines a series of standard protocols.

Application Layer	Telnet	FTP	TFTP	SNMP
	HTTP	SMTP	DNS	DHCP
Transport Layer	TCP		UDP	
Network Layer	ICMP		IGMP	
	IP			
Data Link Layer	PPPoE			
	Ethernet		PPP	
Physical Layer	...			

Common Protocol Standardization Organization

Internet Engineering Task Force (IETF)

- IETF is a voluntary organization responsible for developing and promoting Internet protocols (especially protocols that constitute the TCP/IP protocol suite), and releasing new or replacing old protocol standards through RFCs.

Institute of Electrical and Electronics Engineers (IEEE)

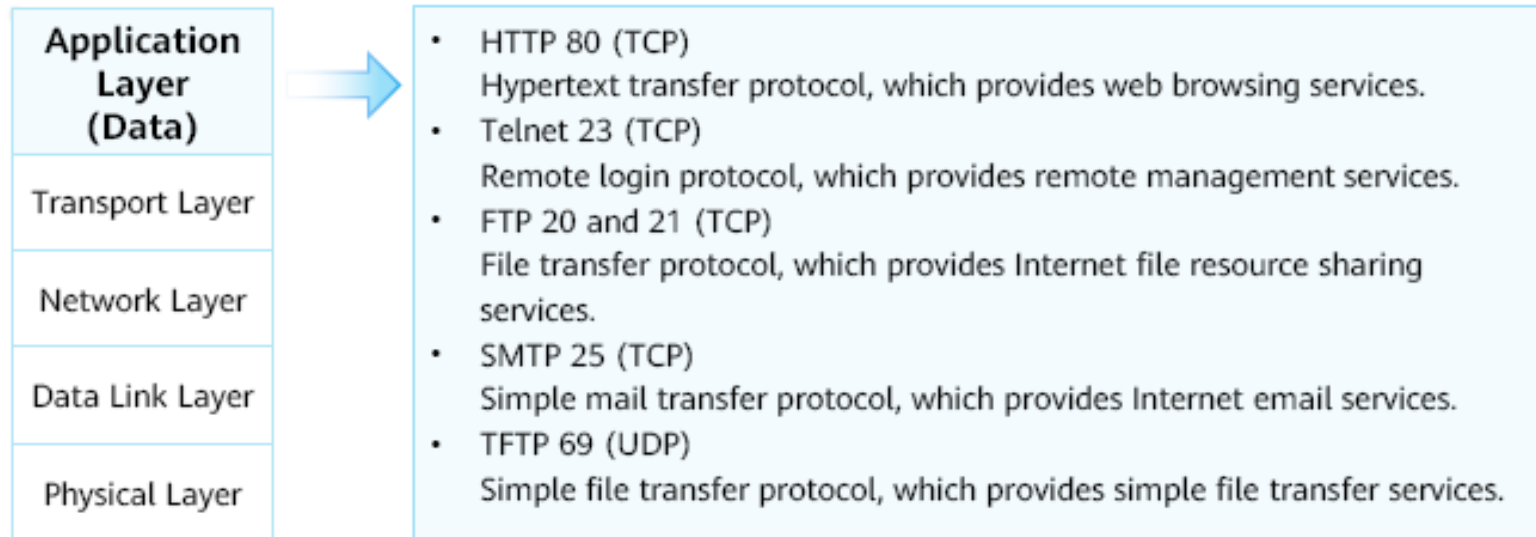
- IEEE has formulated about 30% of standards in the electronics, electrical, and computer science fields worldwide. Those standards include well-known IEEE802.3 (Ethernet) and IEEE802.11 (Wi-Fi).

International Organization for Standardization (ISO)

- ISO is an international organization that plays an important role in the formulation of computer network standards, such as the OSI model defined in ISO/IEC 7498-1.

Application layer

- The application layer provides interfaces for application software so that applications can use network services. The application layer protocol designates transport layer protocols and ports.
- PDUs transmitted at the application layer are called data.



Common Application Layer Protocols - FTP

- The File Transfer Protocol (FTP) transfers files from one host to another to implement file download and upload. This protocol adopts the client/server (C/S) structure.

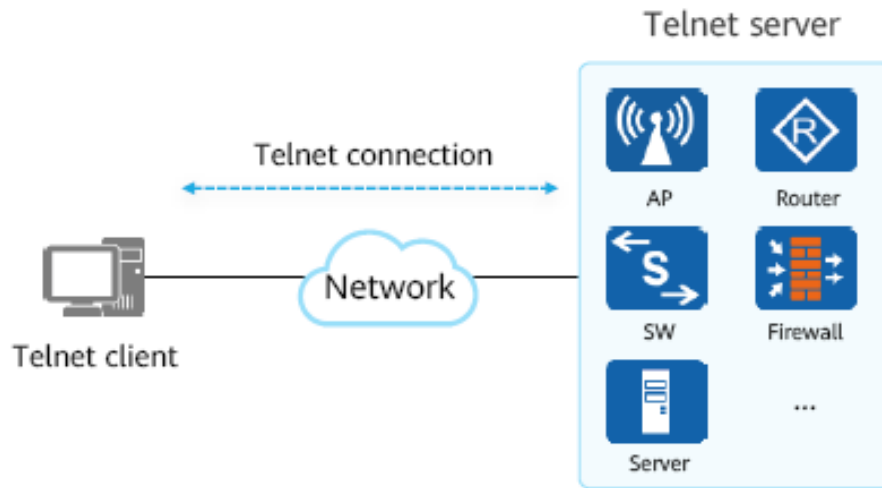


FTP client: provides commands for local users to operate files on a remote server. A user can install an FTP client program on a PC and set up a connection with an FTP server to operate files on the server.

FTP server: a device that runs the FTP service. It provides the access and operation functions for remote clients, allowing users to access the FTP server through the FTP client program and access files on the server.

Common Application Layer Protocols - Telnet

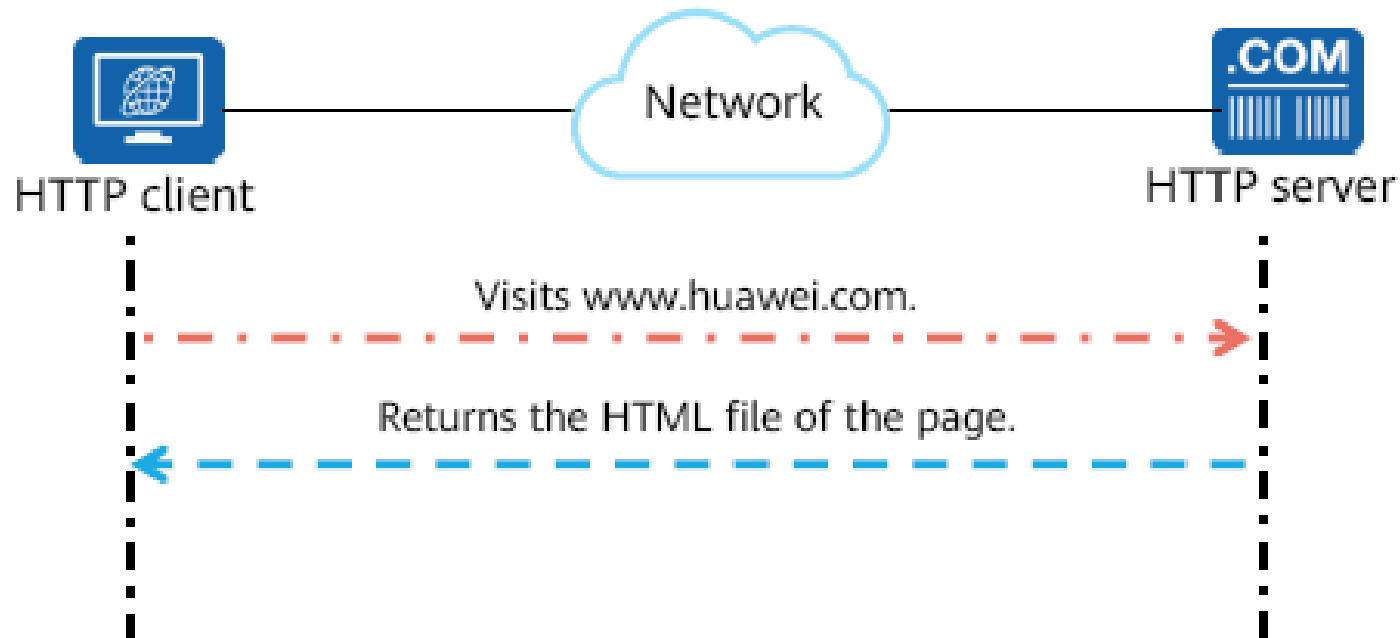
- Telnet is a standard protocol that provides remote login services on a network. It provides users with the ability to operate remote devices through local PCs.



A user connects to a Telnet server through the Telnet client program. The commands entered on the Telnet client are executed on the server, as if the commands were entered on the console of the server.

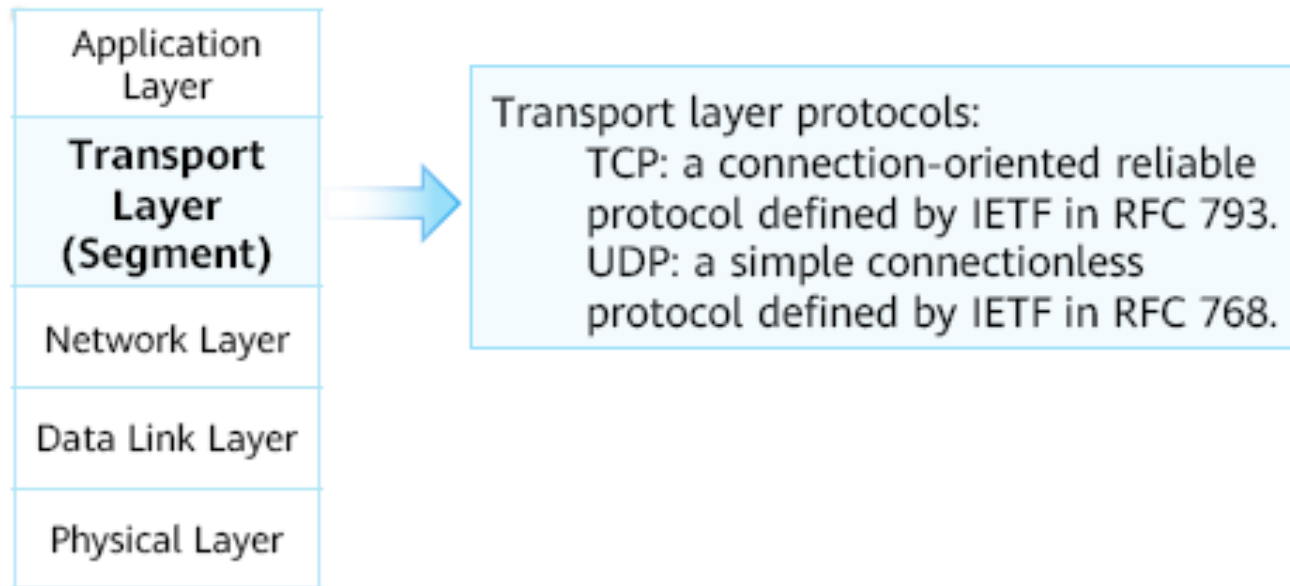
Common Application Layer Protocols - HTTP

- Hypertext Transfer Protocol (HTTP): is one of the most widely used network protocols on the Internet. HTTP was originally designed to provide a method for publishing and receiving HTML pages.



Transport Layer

- A transport layer protocol receives data from an application layer protocol, encapsulates the data with the corresponding transport layer protocol header, and helps establish an end-to-end (port-to-port) connection.
- PDUs transmitted at the transport layer are called segments.

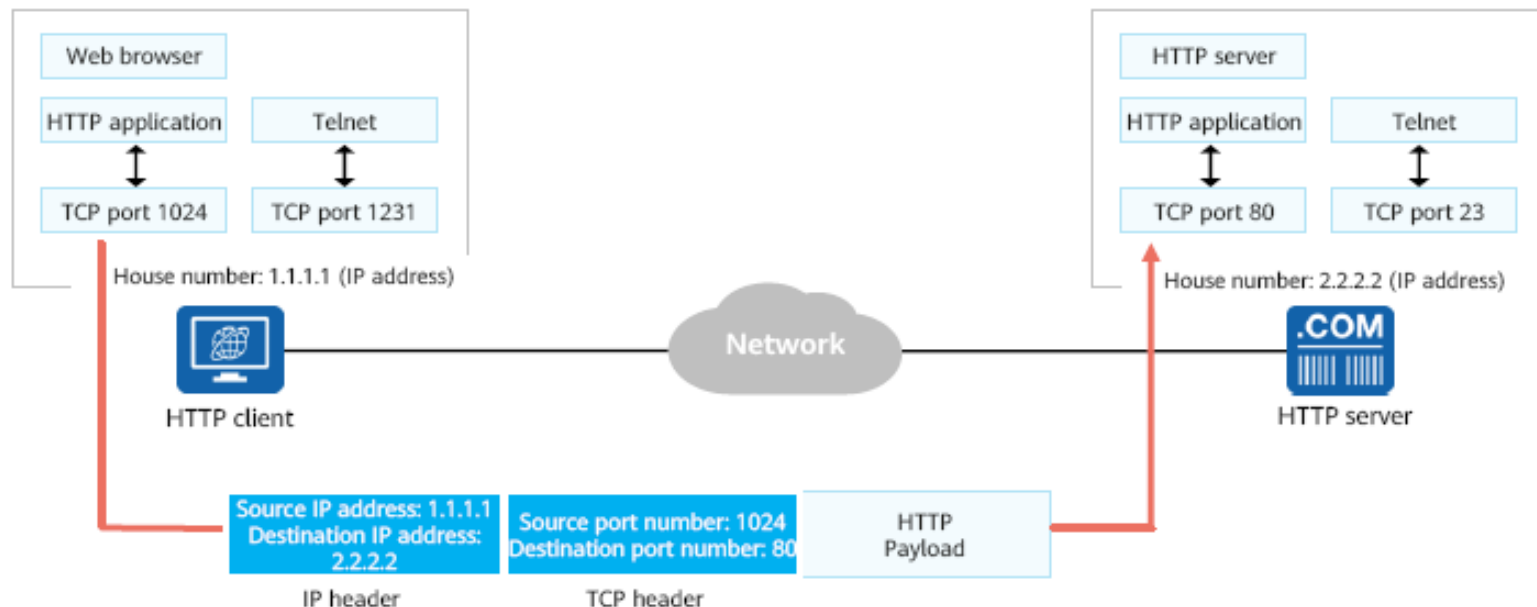


TCP and UDP - Header Formats

Source Port (16)		Destination Port (16)		TCP header 20 bytes
Sequence Number (32)				
Acknowledgement Number (32)				
Header Length (4)	Reserved (3)	Control Bits (9)	Window (16)	
Checksum (16)		Urgent (16)		
Options				
Data (varies)				

Source Port (16)	Destination Port (16)	} UDP header 8 bytes
Length (16)	Checksum (16)	
Data (if any)		

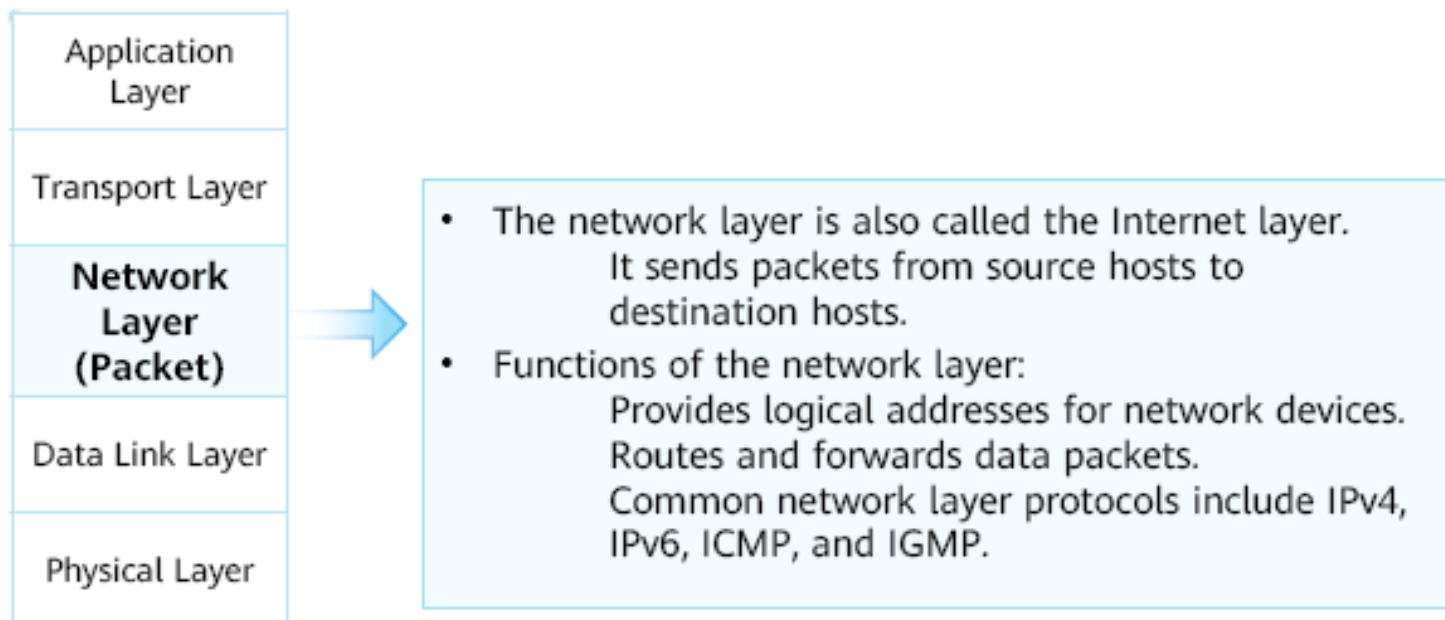
TCP and UDP - Port Numbers



- Generally, the source port used by a client is randomly allocated, and the destination port is specified by the application of a server.
- The system generally selects a source port number that is greater than 1023 and is not being used.
- The destination port number is the listening port of the application (service) enabled on the server. For example, the default port number for HTTP is 80.

Network Layer

- The transport layer is responsible for establishing connections between processes on hosts, and the network layer is responsible for transmitting data from one host to another.
- PDUs transmitted at the network layer are called packets



Working Process of a Network Layer Protocol

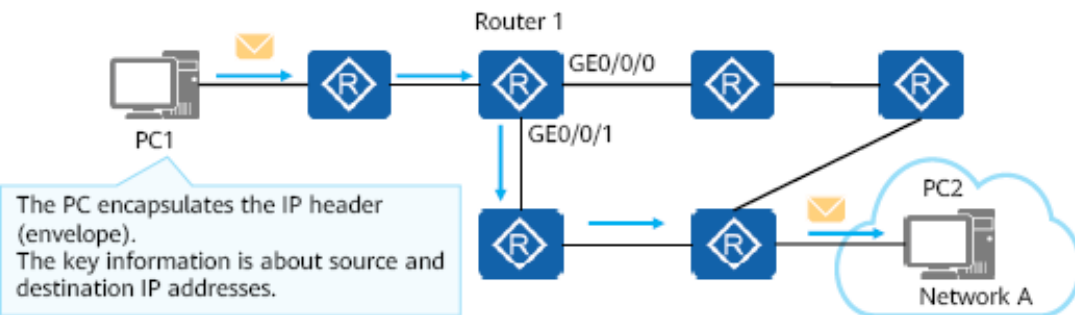
Packet Encapsulation

Letter: data sent by an upper layer (for example, the transport layer)



Envelope: IP packet header
Sender: source IP address
Receiver: destination IP address

Packet Forwarding Based on Network Layer Addresses



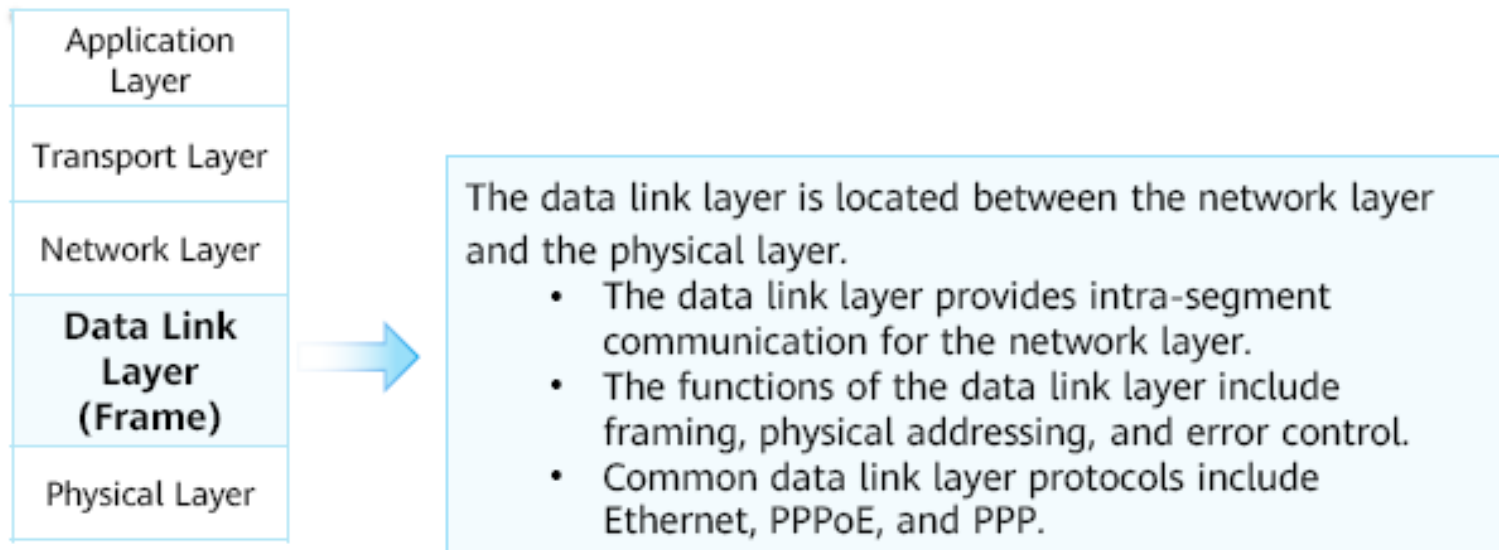
Routing table of Router 1

Network	Outbound Interface
Network A	GE0/0/1
...	...
...	...

- The network layer header of a packet sent by a source device carries the network layer addresses of the source and destination devices.
- Each network device (such as a router) that has the routing function maintains a **routing table** (like a map of the network device).
- After receiving a packet, the network device reads the network layer **destination address** of the packet, searches the routing table for the matching entry of the destination address, and forwards the packet according to the instruction of the matching entry.

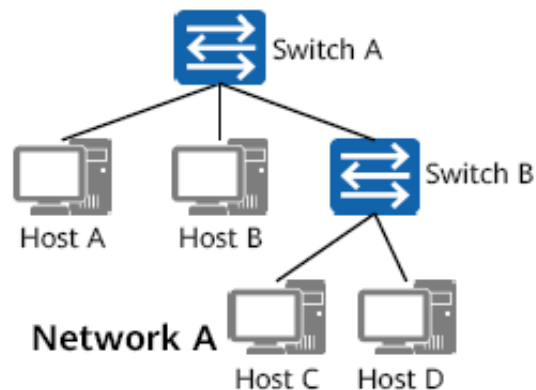
Data Link layer

- The data link layer is located between the network layer and the physical layer and provides services for protocols such as IP and IPv6 at the network layer. PDUs transmitted at the data link layer are called frames.
- Ethernet is the most common data link layer protocol.



Ethernet and Source MAC Addresses

Ethernet Definition



- Ethernet is a broadcast multiple access protocol that works at the data link layer protocol.
- The network interfaces of PCs comply with the Ethernet standard.
- Generally, a broadcast domain corresponds to an IP network segment.

Ethernet Source MAC Addresses

I have a MAC address when I leave the factory.



Host A

Name: Host A



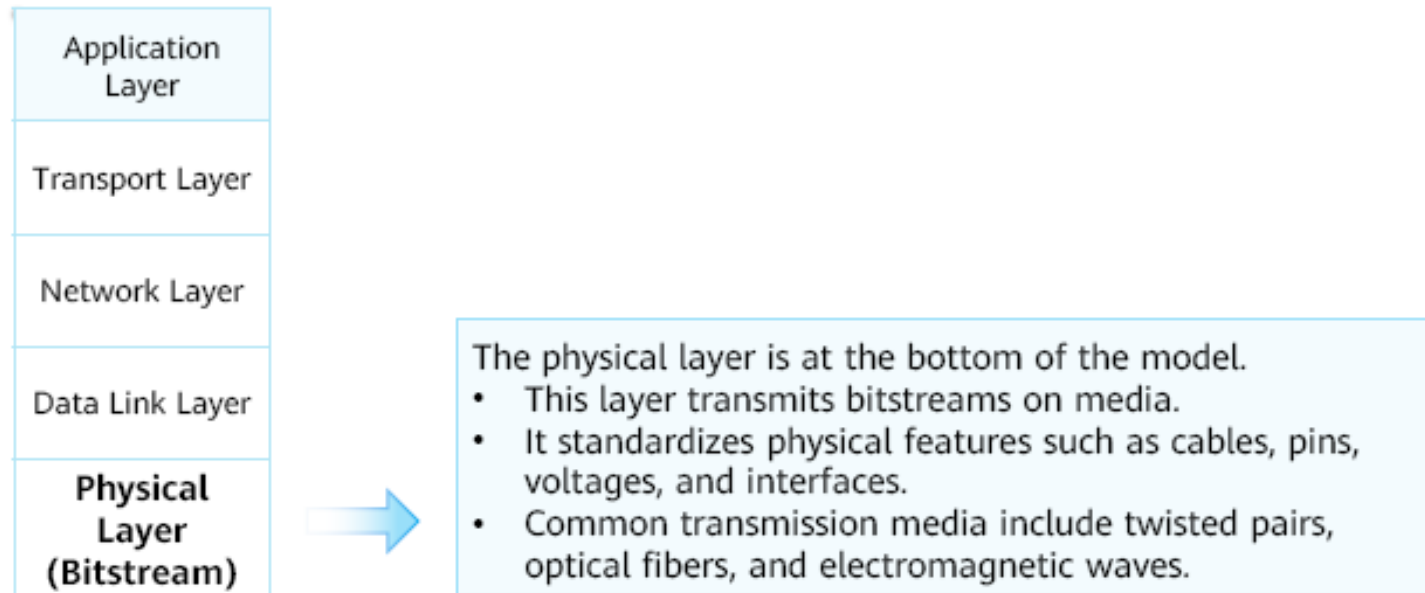
MAC address/Ethernet address/physical address:

--	--	--	--	--	--

- A media access control (MAC) address uniquely identifies a NIC on a network. Each NIC requires and has a unique MAC address.
- MAC addresses are used to locate specific physical devices in an IP network segment.
- A device that works at the data link layer, such as an Ethernet switch, maintains a MAC address table to guide data frame forwarding.

Physical Layer

- After data arrives at the physical layer, the physical layer converts a digital signal into an optical signal, an electrical signal, or an electromagnetic wave signal based on the physical media.
- PDUs transmitted at the physical layer are called bitstreams.



Common Transmission Media



Twisted pair



RJ45 connector

Data transmission through twisted pairs

1



Fiber



Optical module



Data transmission through optical fibers

2



Synchronous/asynchronous
serial cable: V.24 on the left
and V.35 on the right

Data transmission through serial cables

3

4



PAD



Mobile phone



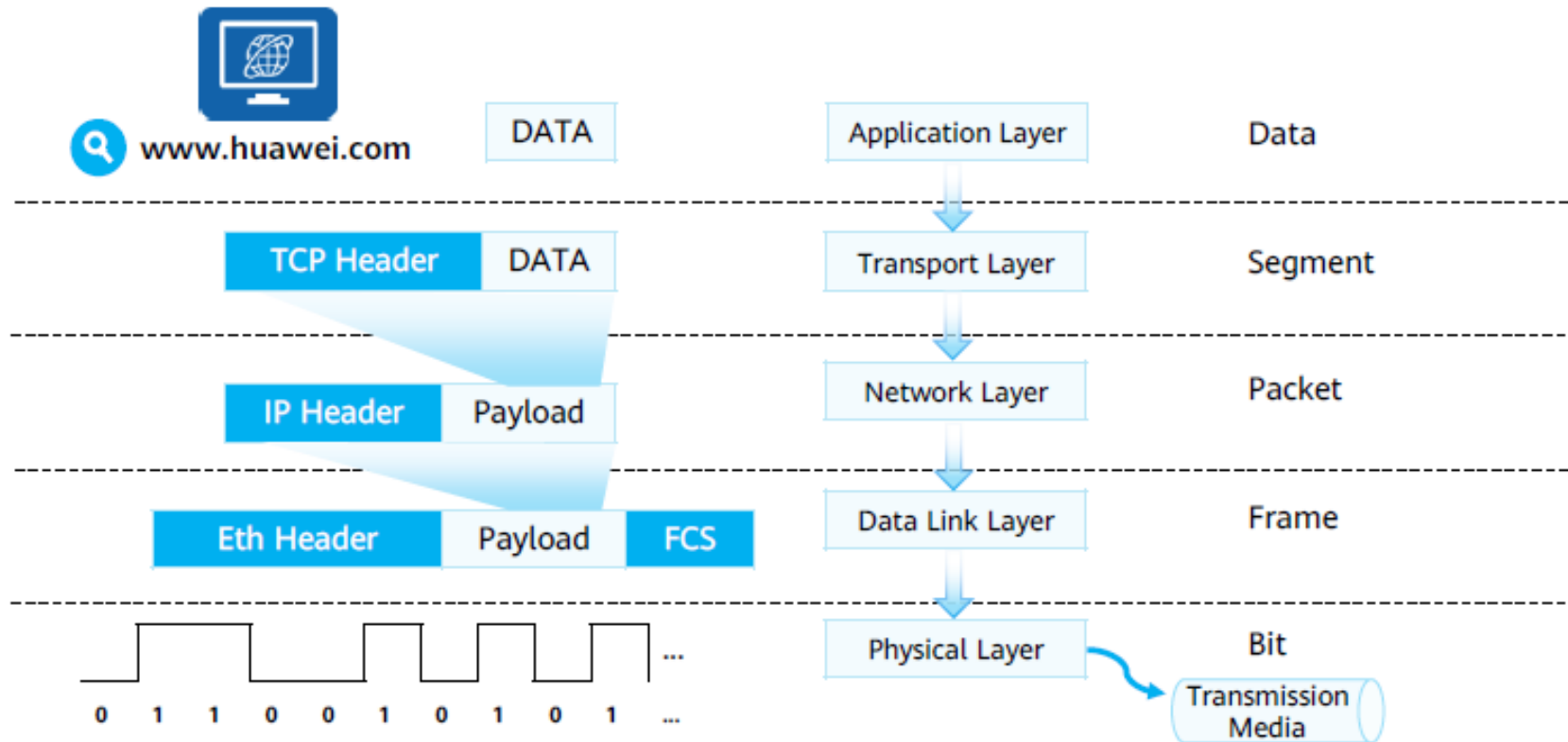
Laptop



Wireless router

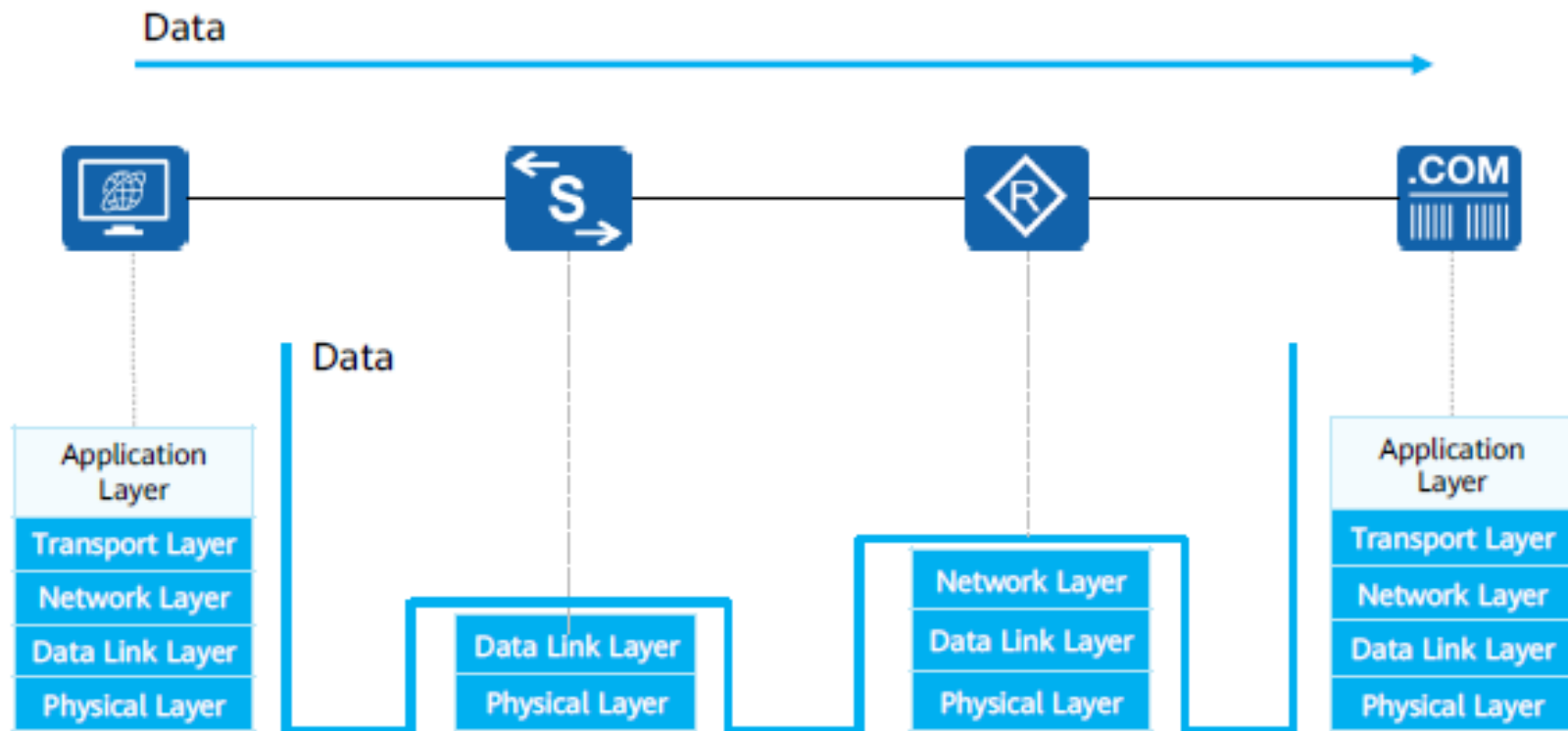
Data transmission between terminal and
wireless routers through wireless signals

Data Encapsulation on the Sender



Data Transmission on the Intermediate Network

- Encapsulated data is transmitted on the network.



Protocolo

- Conjunto de reglas que rigen el intercambio de mensajes entre dos entidades que se comunican para lograr un fin
- Ejemplo, pedir la horas
- Finalidad, obtener información sobre la hora actual de la otra entidad

